Vincent Feliciano

Professor Lin

Cancer Immunotherapy

14 August, 2026

Immunotherapy: PD-1, Immune Evasion, and the Importance of Pembrolizumab

Cancer begins to emerge when a normal cell undergoes genetic damage that affects the proteins it produces, leading to abnormal cell growth, division, and survival. DNA essentially controls protein production, so when mutations occur in our DNA, they can result in incomplete proteins that negatively impact cellular processes. DNA is affected by both nature and nurture, a topic that is often controversial, especially in psychology. Biological causes of DNA damage include errors in DNA replication and the activity of APOBEC enzymes, a family of proteins that our cells use to protect against viruses. They attack viruses by introducing mutations in their DNA. Environmental agents that can cause DNA mutations include smoking, excessive sun exposure, alcohol, an unbalanced diet, and more. Inherited mutations, especially those that affect the BRCA1 and BRCA2 genes, increase the risk of cancer. The BRCA1 and BRCA2 genes help suppress cancer growth. There are two types of mutations: driver and passenger mutations. Driver mutations alter cellular pathways, while passenger mutations are incidental and have no significant impact on a tumor's behavior. Scientists tend to focus more on driver mutations because they provide greater insight into the development of modern-day medicines and therapies. The study of tumors and their evolution has paved the path for the development of modern-day immunotherapies that attack cancer cells without causing severe damage to healthy cells.

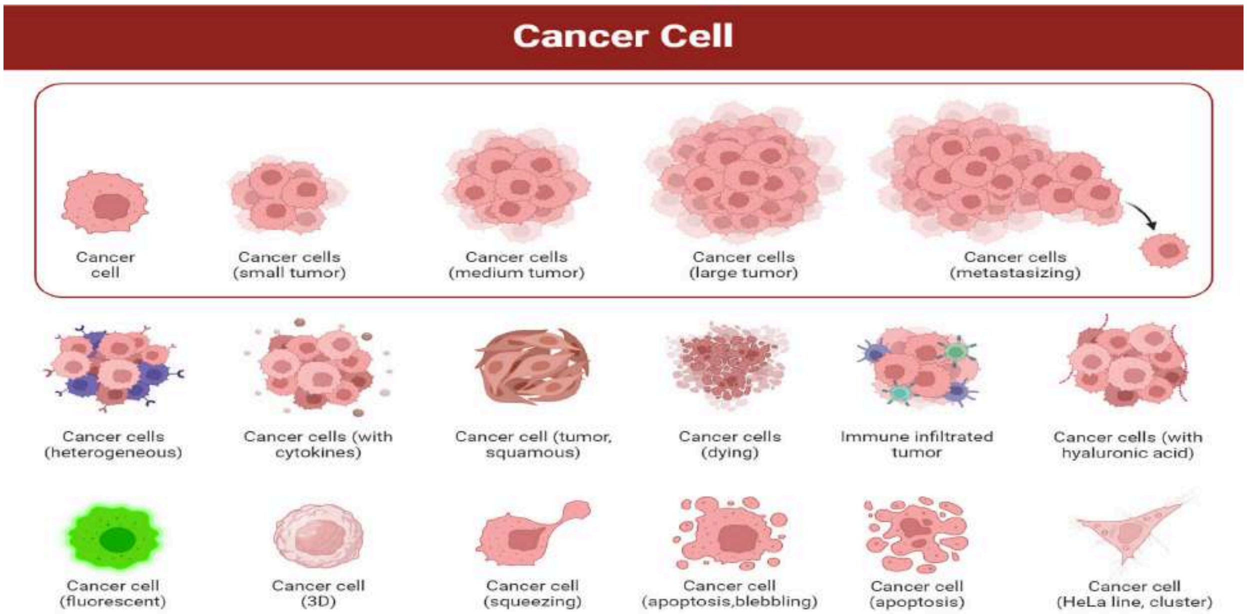

Figure 1. Cancer Cell Evolution: This image depicts how a single cancer cell can proliferate and give rise to larger tumors. It also portrays many forms and behaviors of cancer cells, including metastasis, apoptosis, and tumors infiltrated by immune cells.

In simple terms, immunotherapy is essentially using one's own immune system to fight cancer. Depending on a patient's circumstances, it may be used as a primary treatment or combined with other methods such as chemotherapy. The immune system has natural mechanisms ("checkpoints") that prevent it from becoming too strong and attacking healthy cells and tissue. Some cancers take over these checkpoints and use them to prevent the immune system from attacking the cancer cells. There are many types of immunotherapy methods, but the main three are immune stimulants, T-cell therapy, and monoclonal antibodies. Immune stimulants stimulate the immune system by proliferation or the activation of T-cells and/or natural killer cells. T-cells and natural killer cells are significant in one's immune system. T-cells help a person's immune system adapt to a new disease and attack specific antigens. At the same time, natural killer cells have a spontaneous, innate response, targeting diseases such as cancer without prior knowledge of the matter.

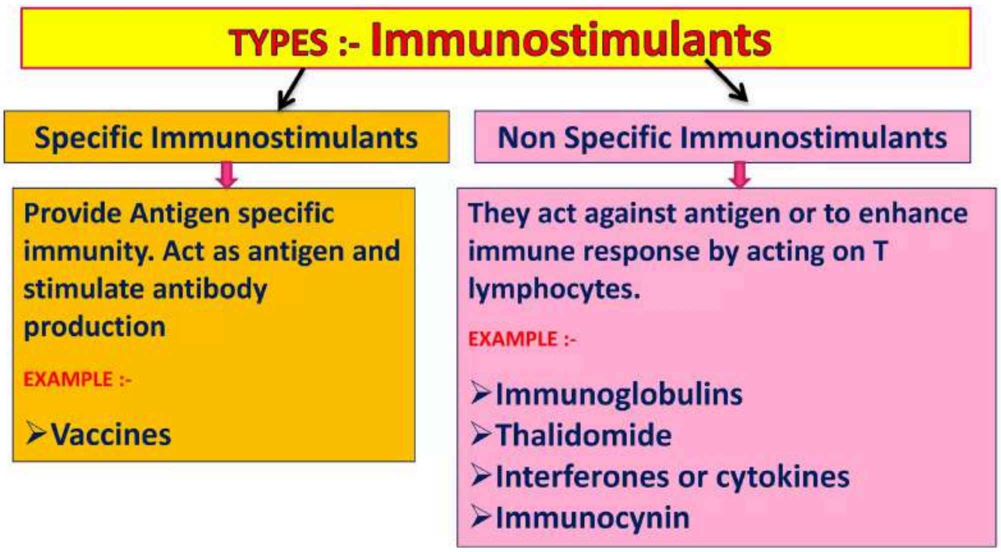

Figure 2. Immunostimulants: This diagram shows differences between specific and nonspecific immunostimulants and their functions. Specific immunostimulants target specific antigens, whereas nonspecific immunostimulants enhance immune responses through mechanisms such as T-cell activation.

T-cell therapy involves scientists taking cells from a patient and engineering the cells to recognize and attack cancer cells more efficiently. Patients then undergo autotransfusion to obtain these engineered cells.

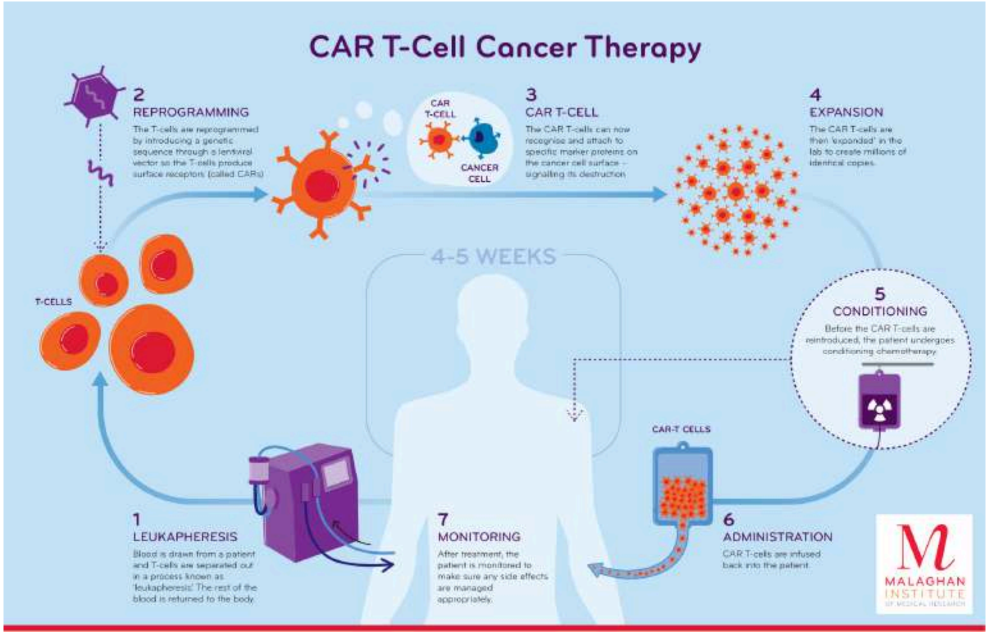

Figure 3. T-Cell Therapy: This diagram shows a step-by-step illustration of CAR T-cell therapy, starting with the removal of a patient's T-cells and followed by the ability to recognize cancer cells. The engineered T-cells are then multiplied and reinfused back into the patient, where they can recognize and attack cancer cells.

Immune stimulants and T-cell therapy have had a great impact on cancer treatment, but I will be focusing on monoclonal antibodies. An antibody is simply a protein that is composed of amino acids. Monoclonal antibodies are lab-manufactured proteins that bind to antigens. An antigen is any foreign substance that elicits an immune response. Checkpoint inhibitors are a type of monoclonal antibody that essentially reactivate T-cells, permitting them to continue attacking cancer cells. The study of the PD-1 pathway has led to the industrialization of checkpoint inhibitors, designed to eliminate PD-1 and PD-L1 receptors.

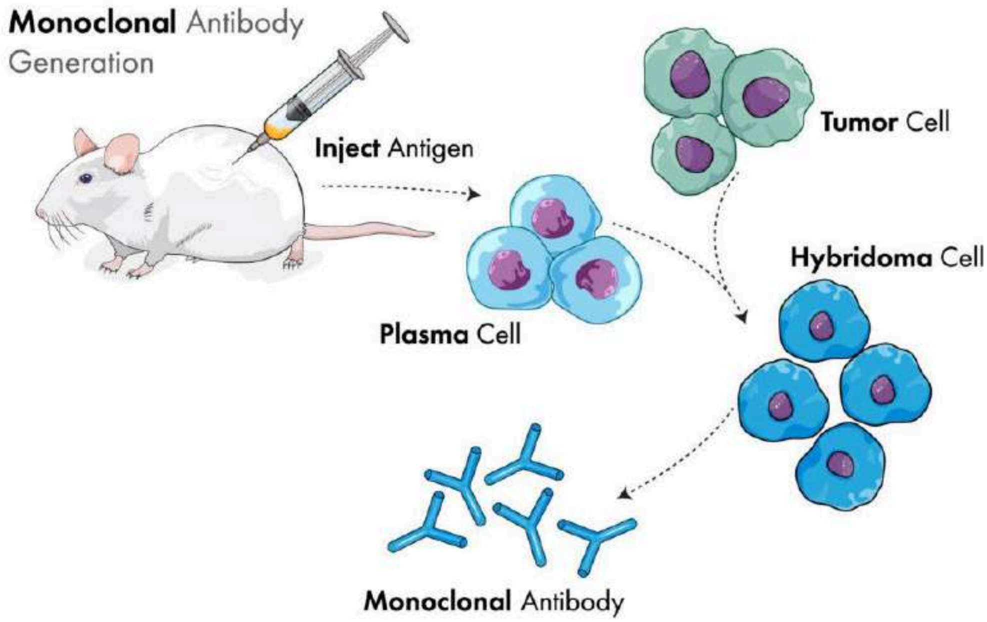

Figure 4. Monoclonal Antibody Process: This image showcases the steps that lead to the production of monoclonal antibodies, beginning with exposure to an antigen and the creation of antibody-producing cells. These antibodies are engineered to recognize and bind to specific antigens, allowing them to attack cancer cells.

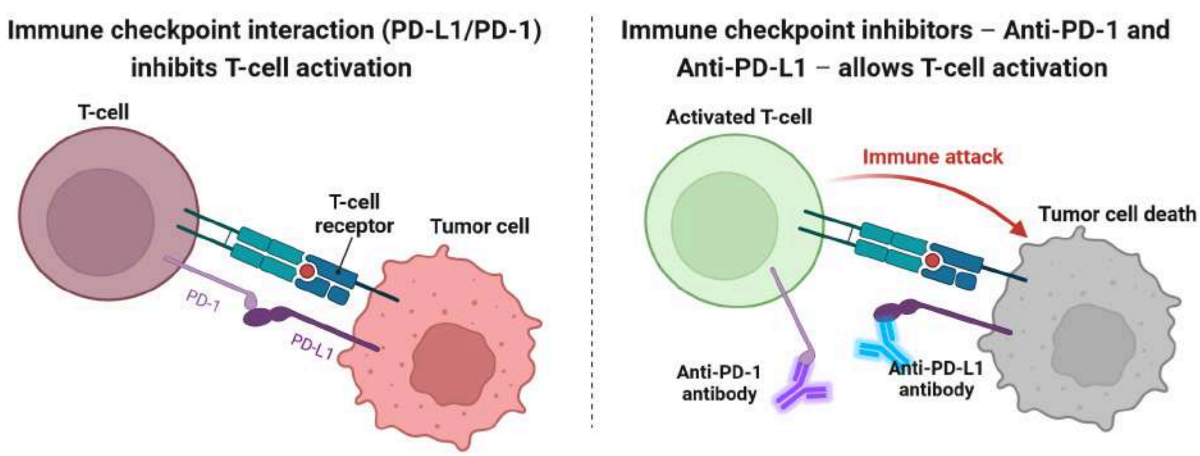

Figure 5. Checkpoint Inhibitors: This diagram shows the difference between suppression of T-cell activation through PD-1/PD-L1 binding and how checkpoint inhibitors prevent

this binding. By blocking PD-1 or PD-L1, T cells can remain active and perform their ordinary function of attacking cancer cells.

Now what is the PD-1 Pathway? PD-1 stands for Programmed Cell Death Protein 1, which is an immune checkpoint pathway that is one of the body's natural mechanisms for reducing excessive immune responses. It is a receptor found mainly on activated T-cells, which attack abnormal cells, such as cancer cells. Under ordinary circumstances, PD-1 binds to its ligands, PD-L1 and PD-L2, to reduce T-cell activity once an immune response is successful. A ligand is a small molecule that binds itself to a larger molecule, usually a protein, to bring about a specific biological effect. Over time, certain cancers have learned to avoid this defensive mechanism. Tumor cells often produce large amounts of PD-L1, which they bind to PD-1 receptors on T-cells. Because of this constant binding, T-cell activation and cell proliferation are reduced. The immune cells get tired and have trouble recognizing cancer cells. This is known as immune evasion, in which tumors continue to grow because the body is unable to recognize their presence in the first place.

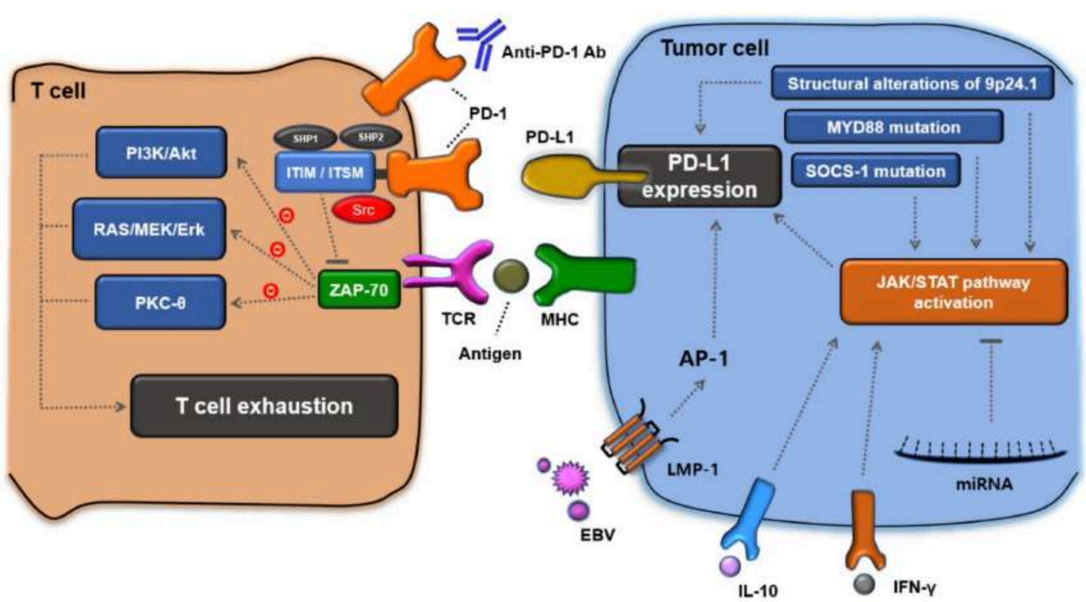

Figure 6. PD-1/PD-L1 and Immune Evasion: This diagram illustrates the molecular interactions between a T-cell and a tumor cell, focusing on the PD-1/PD-L1 checkpoint pathway. Increased PD-L1 expression can hinder T-cell activity, while anti-PD-1 antibodies can reduce this interaction, thereby helping restore the immune system's ability to respond.

Checkpoint inhibitors essentially reactivate T cells, permitting them to continue attacking cancer cells. Unlike chemotherapy, which uses radiation that negatively affects healthy cells, checkpoint inhibitors such as PD-1 attack cancer by using a person's own immune system! The PD-1 immune checkpoint pathway has had a tremendous impact on modern-day oncology, helping scientists understand why PD-1 began to fail when in contact with developing tumors and enabling them to develop checkpoint inhibitors.

The checkpoint inhibitor pembrolizumab, known as Keytruda, is the one I will discuss.

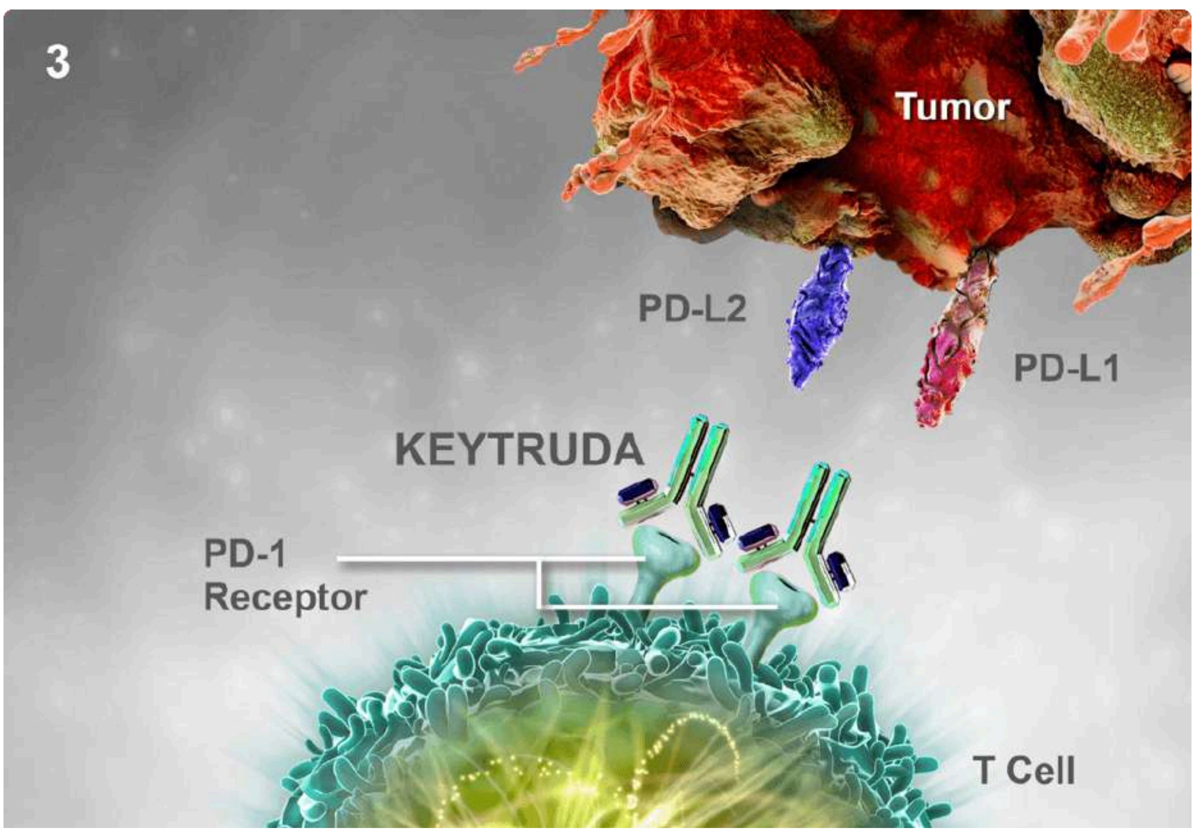

Figure 7. Pembrolizumab/Keytruda: This image illustrates how pembrolizumab (Keytruda) binds to the PD-1 receptor on T-cells, impeding the activation of tumor PD-L1 and PD-L2 checkpoints. By reducing this interaction, pembrolizumab allows T-cells to stay active, improving their ability to identify and eliminate cancer cells.

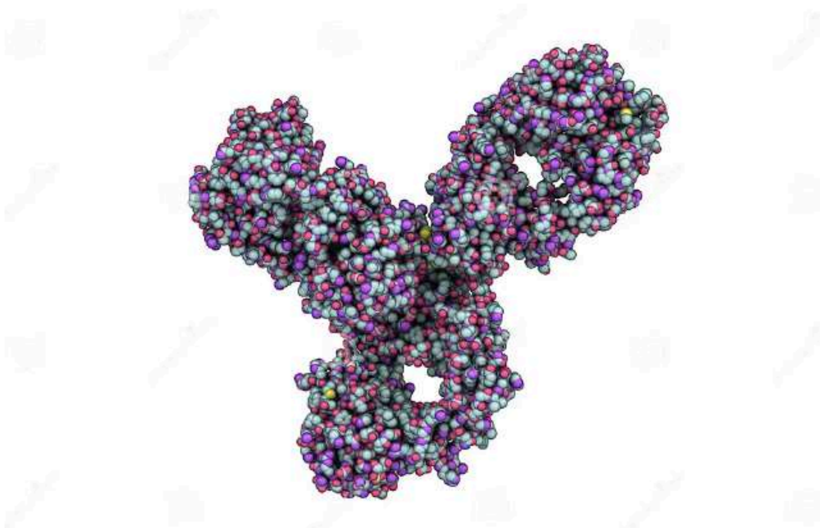

Figure 8. Molecular Structure of Pembrolizumab: The image above illustrates a three-dimensional molecular configuration of pembrolizumab, a monoclonal antibody (checkpoint inhibitor). This configuration allows it to bind to the PD-1 receptor, blocking signals that would otherwise decrease T-cell activity.

The prefix mab at the end of pembrolizumab stands for monoclonal antibody. Keytruda currently has the most MERC out of any other checkpoint inhibitor. MERC stands for Medical Education Research Certificate, meaning it is the most approved by the FDA. Pembrolizumab has had a major impact on NSCLC specifically (non-small cell lung cancer). Pembrolizumab allows T-cells to attack cancer more quickly and effectively. It has become the standard treatment for those who have non-small cell lung cancer. Depending on the patient's condition, it may be combined with chemotherapy or provided solo. Patients whose tumors hold an excessive amount of PD-L1 often receive pembrolizumab first. Many trials over the years have shown that pembrolizumab is an effective checkpoint inhibitor that has saved many lives. Pembrolizumab has also been shown to be better in the long run. Chemotherapy rapidly eliminates cancer cells but also tends to harm healthy cells, whereas pembrolizumab strengthens the immune system. This strengthened immune system recognizes cancer cells even years after treatment, decreasing the risk of redeveloping cancer. Pembrolizumab has immensely altered how scientists perceive cancer treatments and checkpoint inhibitors. Pembrolizumab is a clear demonstration of the effectiveness of immunotherapy and its benefits in cancer.